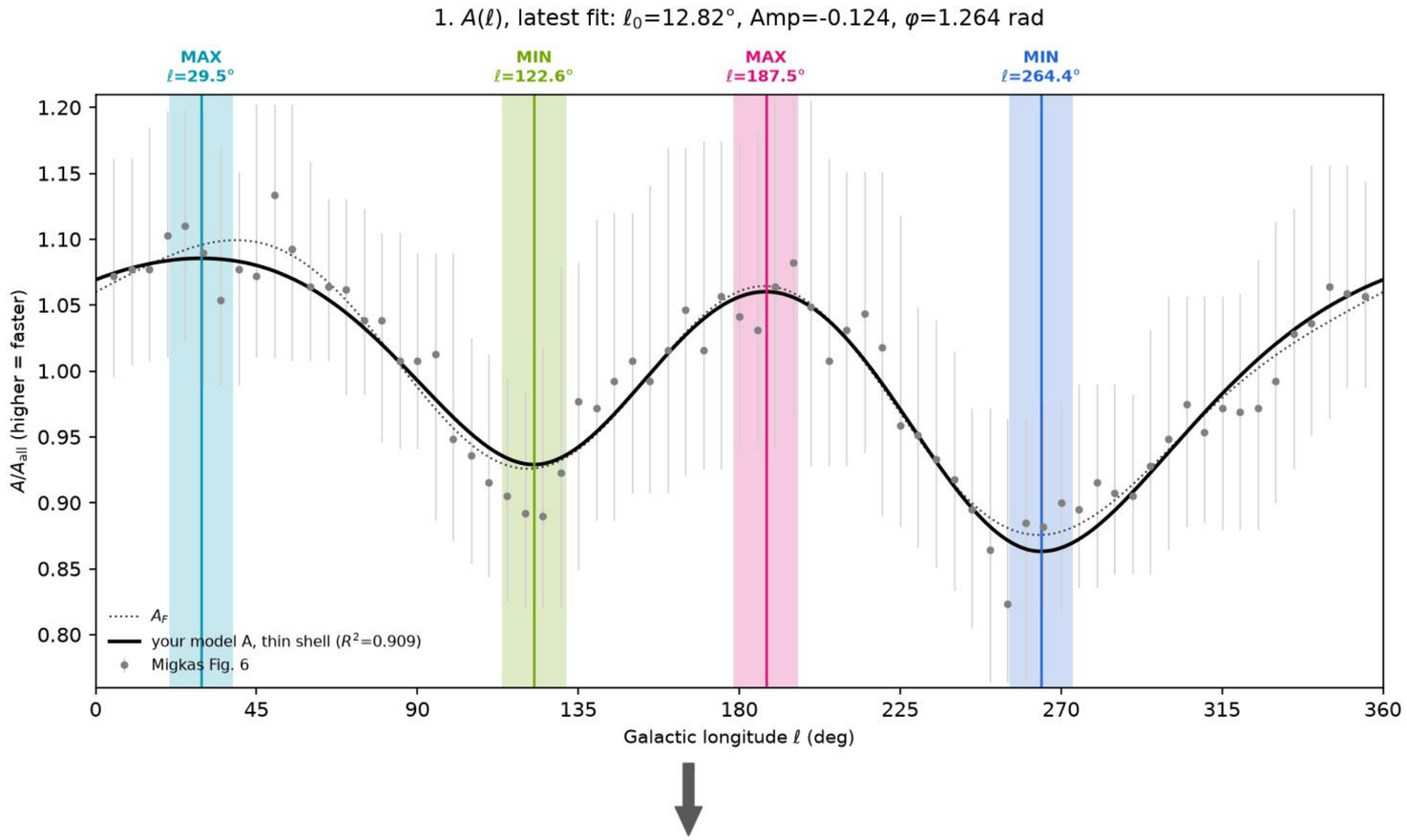
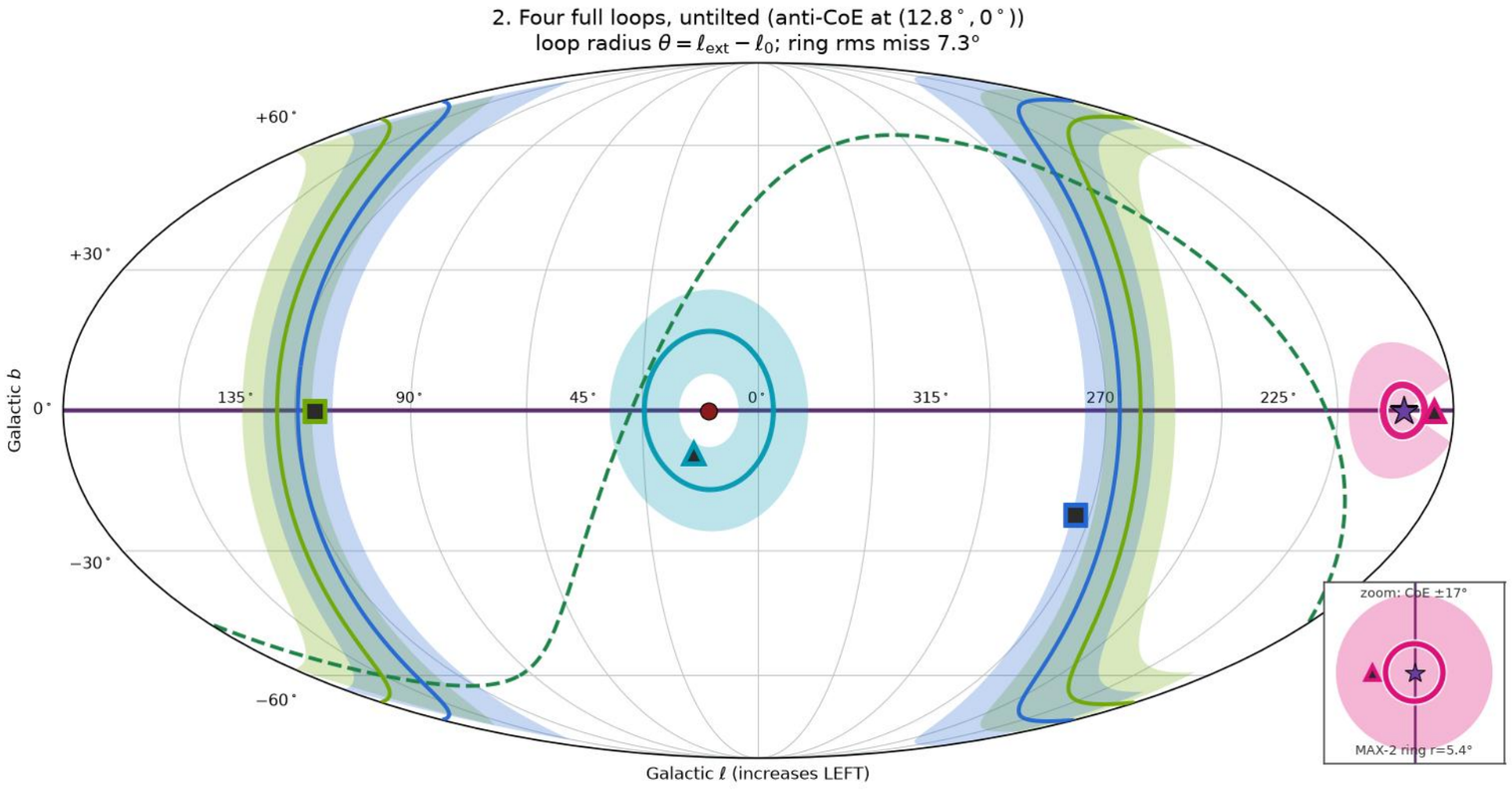
Stacked transformations: from your model A(ℓ) to loops on the sky and the Migkas Fig. 8 overlay
(thin-shell best fit; Galactic Mollweide, ℓ increases LEFT)





Step 1. Gravity breaks homology: your model A

A pure homologous blast (V = H R) would look isotropic from anywhere, so the CoE would drop out. Gravity decelerates matter at different positions relative to the CoE by different amounts, so the expansion rate seen from Earth depends on direction. Your model A (thin-shell best fit) turns this into A(ℓ) ∝ H0² along the Galactic plane, fitted to Migkas Fig. 6 with the parameters in Table 1a. Its two maxima and two minima are read off the curve (Table 1b); each coloured band is ±9° about an extremum, and the same colours are used for the loops on every map below.

Table 1a. Model A fit to Migkas Fig. 6

Parameter	Value
ℓ_0 (anti-CoE longitude)	12.82°
Amp	-0.124
φ	1.264 rad
R²	0.909

Table 1b. Extrema of model A

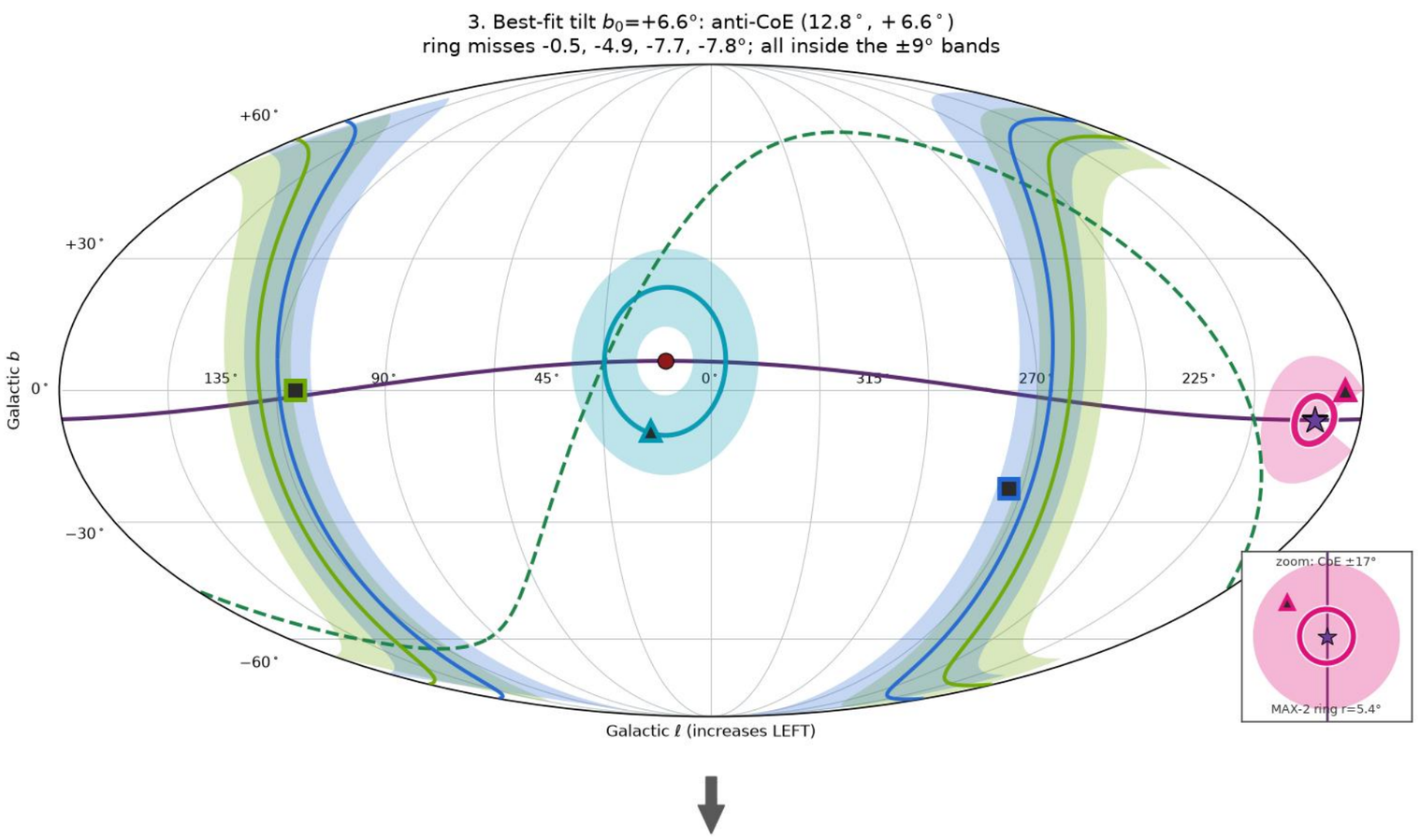
Loop	ℓ_{ext} (°)	A/A_all
MAX-1	29.5	1.085
MIN-1	122.6	0.929
MAX-2	187.5	1.060
MIN-2	264.4	0.863

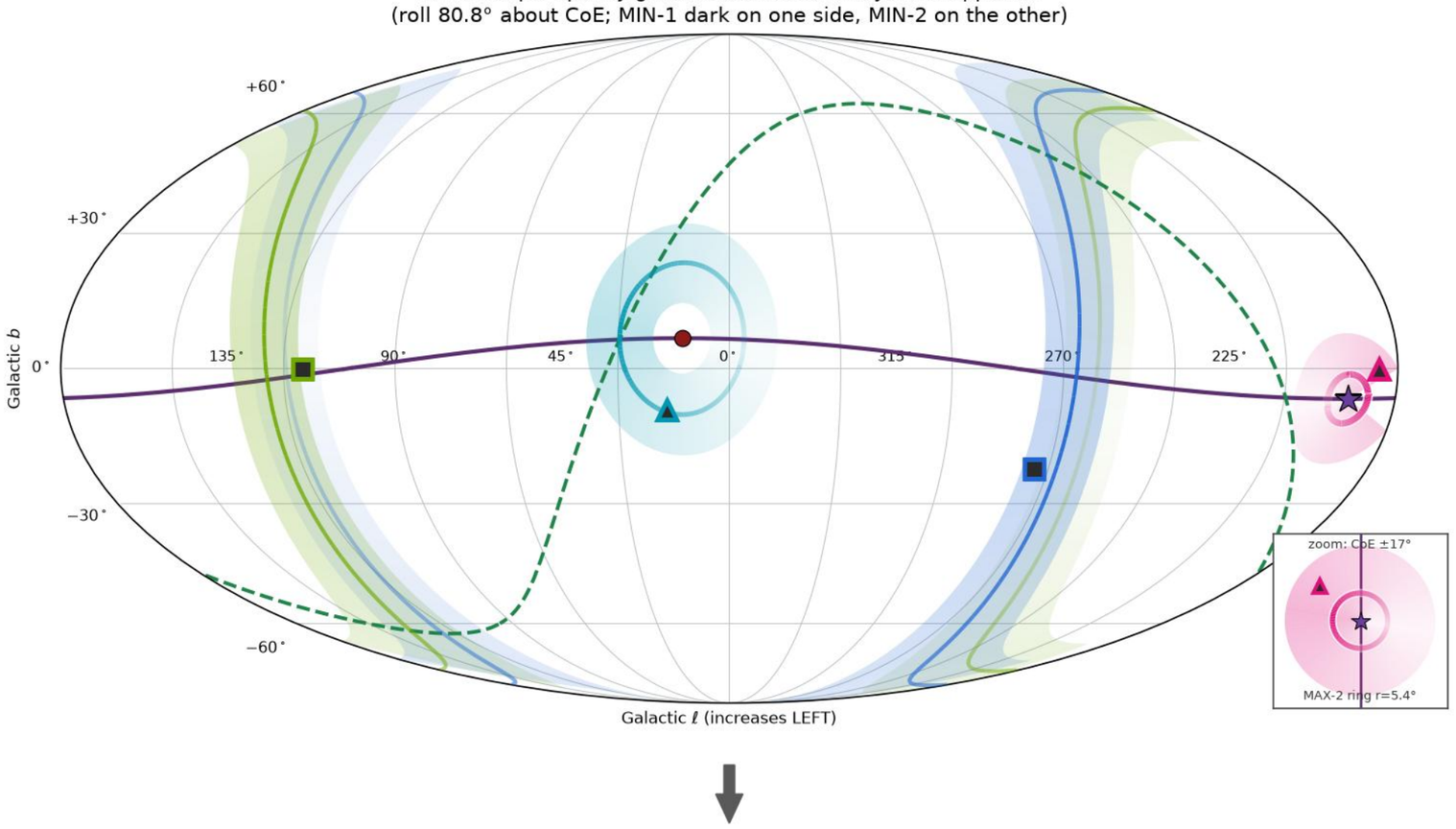
Step 2. Each extremum becomes a loop (untilted)

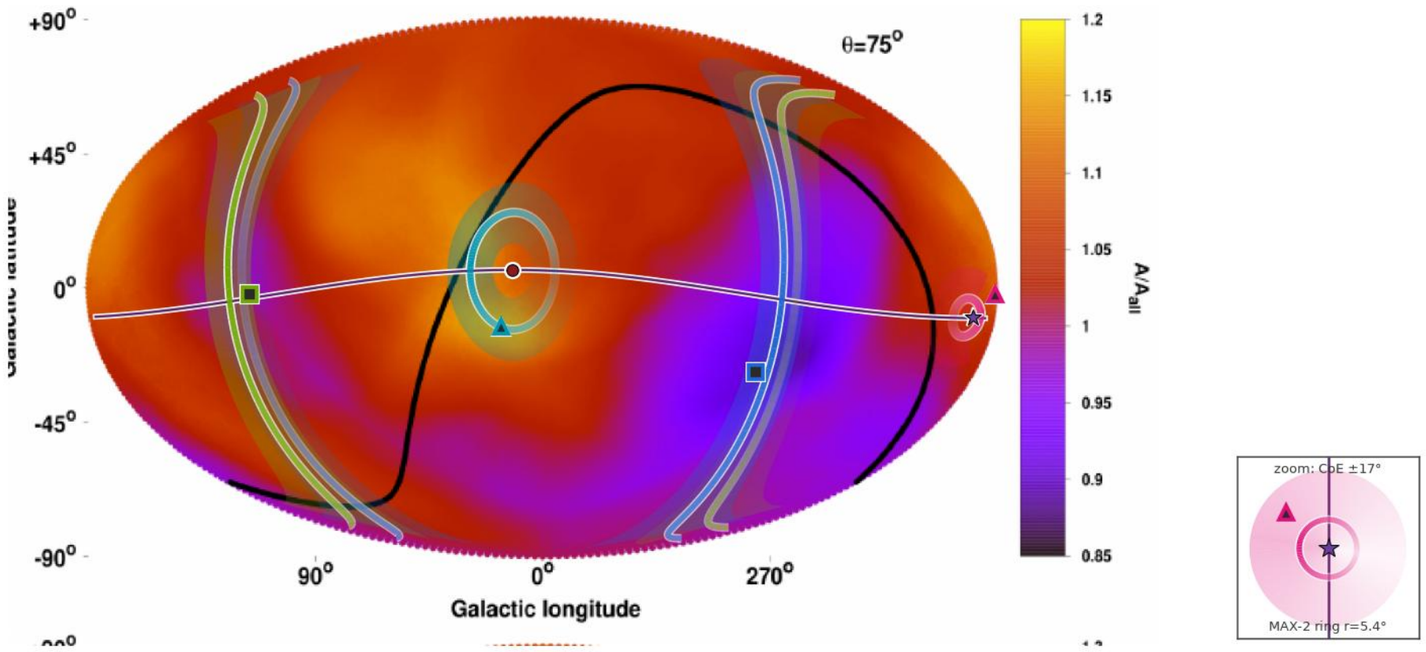
Model A is axisymmetric about the CoE axis, so a value read at longitude ℓ_{ext} in the plane holds on the whole cone of half-angle θ = ℓ_{ext} − ℓ_0 about the anti-CoE direction. Drawing each cone on the sky turns the four extrema into four full loops, here with the axis kept in the Galactic plane (b0 = 0). The ring miss is the signed angular distance from each Migkas extremum to its loop (Table 2). MAX-2 is a small ring 5.4° round the CoE (see inset), and the two MIN loops have almost equal radii, so they are drawn 2° apart for display.

Table 2. Loop radii and ring misses, untilted

Loop	θ (°)	cone radius (°)	Migkas (ℓ, b)	miss (°)
MAX-1	16.7	16.7	(17°, -9°)	-6.8
MIN-1	109.8	109.8	(115°, +0°)	-7.6
MAX-2	174.6	5.4 (CoE)	(185°, +0°)	-2.5
MIN-2	251.6	108.4	(274°, -22°)	-10.2
RMS				7.3







5. Overlay on Migkas Fig. 8, $\theta = 75^\circ$ ($A \propto H_0^2$): four full loops, opacity gradient darkest at θ ray (CoE axis white; celestial equator black)

CoE ↔ anti-CoE axis (great circle)
MAX-1 loop θ=16.7° (ℓ=29.5°), band ±9°
MAX-2 loop θ=174.6° (ℓ=187.5°), band ±9°
MIN-1 loop θ=109.8° (ℓ=122.6°), band ±9° (drawn +2°)
MIN-2 loop θ=251.6° (ℓ=264.4°), band ±9° (drawn +2°)
celestial equator (Dec=0)
Migkas MAX (17°, -9°)
Migkas MAX (185°, +0°)
Migkas MIN (115°, +0°)
Migkas MIN (274°, -22°)
anti-CoE
CoE

Display offset: the two MIN loops (true cone radii 109.8° and 108.4° about anti-CoE) are each drawn 2° further along θ so both lines are seen; all numbers use the true positions. Insets zoom on the 5.4°-radius MAX-2 ring round the CoE.

Step 3. Tilt the CoE axis out of the plane

The anti-CoE longitude is fixed by ℓ_0, but the axis may point out of the Galactic plane. Tilting it to latitude b0 moves all four loops rigidly on the sphere; b0 is fitted by least squares on the ring misses, giving b0 = +6.6° and anti-CoE (12.8°, +6.6°). All four Migkas extrema then fall inside the ±9° bands (Table 3). Freeing ℓ_0 as well only reaches 5.8° RMS, so the remaining ~6° comes from the offset between the model and Migkas extremum longitudes, not from the orientation.

Table 3. Ring misses before and after the tilt (°)

Loop	b0 = 0	b0 = +6.6°
MAX-1	-6.8	-0.5
MIN-1	-7.6	-7.7
MAX-2	-2.5	-4.9
MIN-2	-10.2	-7.8
RMS	7.3	6.0

Step 4. Roll about the CoE axis: the gradient

The tilt fixes where the loops lie but not their orientation about the axis. A roll about the CoE axis sets each loop's θ-ray point, the point where the loop meets the great circle through the axis that plays the role of step 1's slice; there the loop value is the one actually read from A(ℓ). Each loop is drawn darkest at its θ-ray point and fades to 12% opacity opposite, so the dark end marks the measured direction and the faded part is the axisymmetric extrapolation. Because sin θ has opposite signs for MIN-1 and MIN-2, their dark ends lie on opposite sides. The roll, 80.8°, minimises the distance from the θ-ray points to the Migkas extrema (Table 4).

Table 4. θ-ray miss after roll 80.8° (°)

Loop	θ-ray miss
MAX-1	21.9
MIN-1	10.7
MAX-2	6.3
MIN-2	13.9
RMS	14.4

Step 5. Compare with Migkas Fig. 8

Finally the tilted and rolled loops are laid on Migkas et al.'s Fig. 8 map at θ = 75°, with the same registration as your earlier overlays, so they can be compared directly with the observed A/A_all pattern (higher = faster, A ∝ H0²). The white line is the CoE axis great circle and the black line is Migkas's celestial equator. No parameter is fitted in this step; it is the check of steps 1 to 4 against the data.